# ChangeDINO: Bi-Level Feature Entanglement Network for Remote Sensing Change Detection With Vision–Language Prior

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***Abstract*—The performance of most existing remote sensing change detection (RSCD) methods is often constrained by limited semantic understanding and a heavy dependence on large volumes of training data. To address these limitations, we propose ChangeDINO, a novel change detection paradigm that effectively integrates both semantic knowledge and positional information from Grounding DINO, a representative open-vocabulary detector. To fully leverage Grounding DINO's semantic guidance and object localization capabilities, we introduce a bi-level feature entanglement strategy, incorporating two key components: the semantically guided fusion module (SGFM) and the position-aware enhancement module (PEM). The SGFM adaptively exploits the spatially heterogeneous activation patterns of features from the open-vocabulary detector as semantic guidance signals, selectively amplifying the contribution of high-activation object regions while suppressing uninformative background responses. The PEM, in turn, exploits positional embeddings from cross-modal queries and applies feature-wise linear modulation (FiLM) for pixel-level calibration, thereby enhancing spatial precision and refining boundary localization of changed regions. Together, these mechanisms enable ChangeDINO to transfer fine-grained semantics and precise positional cues from large-scale vision–language pretraining into the RSCD task, achieving state-of-the-art (SOTA) performance on standard benchmarks—LEVIR-CD, WHU-CD, and SYSU-CD—as well as on our newly proposed AUUCDD, with $F1$-scores of 92.89%, 95.79%, 83.87%, and 90.17%, respectively. Furthermore, experiments under data-scarce conditions demonstrate that ChangeDINO maintains superior performance by leveraging Grounding DINO's rich pretrained representations, thereby reducing dependence on task-specific annotations and providing a practical solution for data-limited scenarios. The code and datasets are publicly available at https://github.com/G124556/ChangeDINO**

***Index Terms*—Change detection, data-scarce, semantic guidance, vision–language model.**

Received 23 August 2025; revised 13 November 2025, 4 April 2026, 22 May 2026, and 26 June 2026; accepted 14 July 2026. Date of publication 20 July 2026; date of current version 28 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U25A20535, Grant U23B2011, and Grant U2541205; in part by Sichuan Science and Technology Program under Grant 2025YFNZH0034; and in part by the Science and Technology Plan Special Program of Huzhou under Grant 2024ZD2047. *(Corresponding author: Tianyu Li.)*

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Digital Object Identifier 10.1109/TGRS.2026.3715122

## I. Introduction

REMOTE sensing change detection (RSCD) refers to the task of identifying changes in the Earth's surface from bitemporal remote sensing images. By offering valuable insights into dynamic landscape changes over time, RSCD plays a critical role in a wide range of applications, including urban planning, land use monitoring, disaster management, and agriculture [1], [2], [3], [4], [5], [6]. Traditional RSCD methods, such as image differencing and principal component analysis (PCA) [7], [8], [9], primarily rely on rule-based techniques to detect changes. However, these methods struggle to handle complex and heterogeneous landscapes [10] due to their reliance on predefined thresholds and simplistic assumptions.

The rapid development of deep learning in recent years has significantly advanced the field of RSCD. These learning-based methods are recognized for their ability to automatically extract discriminative features and capture complex landscape patterns from training datasets [11], [12], [13]. Among these, Siamese networks have emerged as the dominant structure, utilizing two identical branches to extract temporal features in parallel and perform change detection by comparing the learned representations [14], [15], [16], [17]. Following this structure, numerous approaches have been proposed, with some focusing on the design of backbone networks, others on feature fusion [18], [19], [20], [21], and still others introducing auxiliary tasks to provide additional supervision [22], [23]. Although these methods have achieved success in many change detection applications, they often overlook prior knowledge that could be leveraged and lack semantic-aware attention to specific change targets, leaving room for further enhancement, particularly in situations where training data is insufficient. While data augmentation strategies have demonstrated value in remote sensing image classification through adaptive sample generation [24] and class imbalance handling [25], these approaches primarily address data quantity issues rather than the fundamental lack of semantic understanding in change detection tasks.

Recently, the emergence of vision–language foundation models has demonstrated remarkable potential in remote sensing with their prior knowledge embedded in large-scale pretrained models. For example, SpectralGPT [26] not only enhances the performance of remote sensing image

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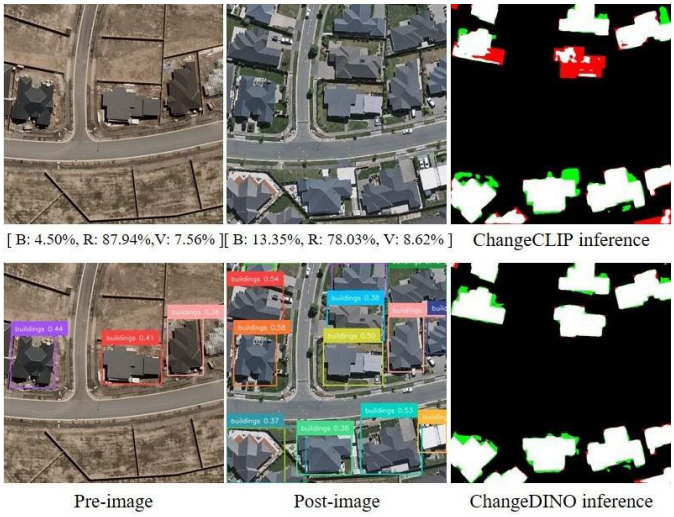



Fig. 1. Comparative analysis of change detection between ChangeCLIP and ChangeDINO: The top row displays bi-temporal images with CLIP scene matching scores (values in brackets denote CLIP's probabilities for scene categories: buildings-B, roads-R, and vegetation-V) and ChangeCLIP's results. The bottom row visualizes Grounding DINO's object detection outcomes on bi-temporal images (showing detection boxes and confidence scores for buildings) alongside ChangeDINO's results. In the third column, white regions indicate true positives, red regions indicate false negatives, while green regions indicate false positives.

classification and segmentation, but also exhibits excellent generalization ability in few-shot scenarios. RemoteCLIP [27] successfully transfers the cross-modal alignment capabilities of CLIP [28] to the remote sensing domain, achieving more accurate scene recognition and object detection. In RSCD, ChangeCLIP [29] improves change detection by integrating scene-level semantic knowledge from the vision–language model CLIP and achieves state-of-the-art (SOTA) performance on multiple public datasets. Although CLIP provides valuable semantic information through image–text pairs, the scene information it offers remains at a global level, as CLIP can only generate coarse scene-level classification probabilities. The first row of Fig. 1 highlights this limitation: when provided with the text array "building, road, vegetation," CLIP predicts the scene contains building with 4.50% confidence, road with 87.94% confidence, and vegetation with 7.56% confidence.

Such coarse understanding provides only limited positional information for localizing change areas, motivating the exploration of open-vocabulary detectors [30] that offer richer object-level priors. Grounding DINO [31] exemplifies such open-vocabulary detectors. As shown in the second row of Fig. 1, when prompted with "buildings," it precisely localizes building instances and provides fine-grained confidence scores. Its feature enhancer produces spatially aligned semantic features via multilayer cross-modal attention, while its language-guided query selection module encodes structured positional information through multiscale feature interaction. These two outputs are highly complementary for change detection. However, directly incorporating them into pixel-level change detection introduces two structural challenges that are not unique to Grounding DINO but are inherent to adapting open-vocabulary detectors to dense prediction tasks.

1) *Spatially Heterogeneous Semantic Activation:* The semantic features exhibit spatially heterogeneous activation patterns—regions corresponding to detected objects produce concentrated, high-magnitude responses, while nontarget regions yield diffuse, low-magnitude activations; indiscriminate fusion inevitably conflates informative object-region semantics with uninformative background activations.
2) *Object-to-Pixel Granularity Gap:* The positional priors are encoded at object granularity—capturing object locations and interobject spatial relationships through language-guided multiscale interactions—which cannot be naively projected onto a spatial grid to yield meaningful pixel-level guidance, as doing so discards the structured nature of these representations and fails to establish per-pixel correspondence.

To address the above challenges, we propose ChangeDINO, a dual-stream change detection framework with two dedicated components. Semantically guided fusion module (SGFM) adaptively exploits spatially heterogeneous activation patterns of the open-vocabulary detector for selective semantic guidance. Position-aware enhancement module (PEM) bridges the object-to-pixel granularity gap by reorganizing positional priors into per-pixel spatial modulation signals via feature-wise linear modulation (FiLM) [32]. Together, these two modules constitute a bi-level feature entanglement mechanism that enables systematic exploitation of Grounding DINO's complementary outputs for pixel-level change detection. The main contributions of this article are as follows.

1) We propose **ChangeDINO**, a novel change detection framework that integrates semantic and positional priors from an open-vocabulary detector (Grounding DINO) into pixel-level change detection via a bi-level feature entanglement mechanism.
2) We propose **SGFM** to address spatially heterogeneous semantic activation by selectively amplifying object-region semantics while suppressing diffuse background responses, and **PEM** to address the object-to-pixel granularity gap by transforming object-level positional queries into per-pixel spatial guidance.
3) We introduce **AUUCDD**, a novel drone-based dataset specifically designed for unauthorized urban construction detection, comprising 2000 high-resolution images ($3024 \times 4032$ pixels) generating 18 000 image pairs, providing a real-world benchmark for evaluating RSCD methods in low-altitude urban monitoring scenarios.
4) ChangeDINO achieves competitive performance on multiple public benchmarks, and demonstrates superior robustness under data-scarce conditions by leveraging rich pretrained representations from the open-vocabulary detector.

## II. Related Works

### A. Siamese Network-Based RSCD

The Siamese network architecture [33] has garnered significant attention for change detection tasks due to its ability to directly capture spatial and contextual dependencies between

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image pairs. A typical Siamese network consists of two identical neural networks that share weights and process two images simultaneously, detecting pixel-wise differences by comparing their learned feature representations. Early Siamese networks primarily rely on simple convolutional neural networks (CNNs) for feature extraction, such as FC-EF [34] and ESCNet [35]. To address the limitations of capturing features at different scales, multiscale feature fusion is adopted in methods like DSIFN [36] and SNUNet [18]. With the advent of Transformer [37] models, the self-attention mechanism is widely incorporated into the Siamese structure to capture long-range dependencies. For example, Zhuo et al. [38] leverage the multihead self-attention mechanism to integrate contextual information, while ChangeFormer [14] adds a cross-scale fusion module alongside self-attention to detect both fine-grained and large-scale changes with high accuracy. To further enhance the efficiency of long-range dependency capture, ChangeMamba [39] integrates state-space models into the Siamese network, achieving competitive performance in change detection tasks. However, these methods face two primary limitations: 1) heavy dependence on large-scale annotated data—their performance significantly degrades when training samples are limited; and 2) lack of high-level semantic understanding—they primarily rely on low-level visual features without incorporating semantic prior knowledge, making them struggle with complex semantic changes.

### B. Vision–Language Models for RSCD

Recent research has explored various approaches to apply multimodal learning in remote sensing. For instance, Sat2Cap [40] proposes a method for generating fine-grained textual descriptions from satellite images, demonstrating the potential of vision–language models in understanding the semantic content of remote sensing scenes. Mendieta et al. [41] investigate how to adapt general-purpose vision–language foundation models to the geospatial domain through continual pretraining, providing stronger semantic representations for remote sensing applications. In addition, Cong et al. [42] explore relational reasoning mechanisms with multiscale attention for remote sensing image understanding, highlighting the importance of semantic feature integration. These works collectively demonstrate that leveraging rich prior knowledge from vision–language pretrained models can effectively enhance the performance of remote sensing image understanding tasks.

Benefiting from large-scale data training and advanced multimodal fusion techniques, vision–language models such as CLIP show powerful semantic understanding capability and remarkable zero-shot adaptivity in remote sensing applications [43], [44], [45]. Applying the multimodal vision–language scheme to RSCD, ChangeCLIP leverages robust semantic information from image–text pairs through its differential feature compensation module and transformer-based multimodal decoder. More recently, MMChange [46] further advances VLM-based change detection by introducing dynamic per-image textual descriptions generated by a generative VLM, representing an effective approach to incorporating large-scale VLM prior knowledge into the change detection task. Together, these works validate the effectiveness of VLM-based semantic priors in enhancing RSCD performance. While these approaches achieve semantic understanding at the image or prompt level, they motivate us to explore open-vocabulary detectors that offer richer pixel-level spatial priors for change detection.

Similar to CLIP, Grounding DINO is also a text-prompted vision–language model. However, unlike CLIP, which achieves image–text alignment solely through global contrastive learning, Grounding DINO employs a more sophisticated internal architecture, mainly including a visual encoder, a text encoder, a feature enhancer, a cross-modal query generator, and an interactive decoder. This enables the model to progress from coarse-grained semantic understanding to fine-grained object localization. Specifically, the feature enhancer achieves fine-grained alignment between visual and language features through multilayer cross-attention mechanisms. When given specific text prompts, the features processed by the feature enhancer retain rich semantic information of the target objects. Meanwhile, the language-guided query selection module generates query vectors through language guidance, which gradually acquire precise spatial positioning information through interaction with multiscale feature maps and positional encoding. The resulting features contain not only accurate target position information but also encode spatial relationships between targets and the surrounding environment. These two types of outputs are potentially valuable for change detection: the semantic features from the feature enhancer facilitate understanding of semantic changes, while the positional features from the language-guided query selection module carry structured spatial information about target locations. However, directly incorporating them into pixel-level change detection introduces two structural challenges inherent to adapting open-vocabulary detectors to dense prediction tasks: spatially heterogeneous semantic activation, where indiscriminate fusion conflates informative object-region semantics with uninformative background activations; and the object-to-pixel granularity gap, where object-level positional representations cannot be naively projected onto a spatial grid to yield meaningful per-pixel guidance. These challenges motivate the design of ChangeDINO, which directly exploits the internal feature-level outputs of the open-vocabulary detector through dedicated adaptation modules.

## III. Method

### A. Overall Architecture

To effectively incorporate prior knowledge from Grounding DINO into the change detection task, we propose the ChangeDINO framework. As illustrated in Fig. 2, this framework adopts a dual-stream architecture that processes bi-temporal images $I^t$, $I^{t+1} \in \mathbb{R}^{3\times H\times W}$ through a weight-shared backbone network, extracting multiscale feature maps $F_i^t$, $F_i^{t+1} \in \mathbb{R}^{C_i\times H/2^{i+1}\times W/2^{i+1}}$, where $i \in \{1, 2, 3, 4\}$ represents different feature levels. $I^t$ and $I^{t+1}$ are also fed into a frozen Grounding DINO model to generate semantic features $G_{s,i}^t$, $G_{s,i}^{t+1}$ and positional features $G_r^t$, $G_r^{t+1}$.

Based on these complementary features, a bi-level feature entanglement mechanism is proposed to systematically

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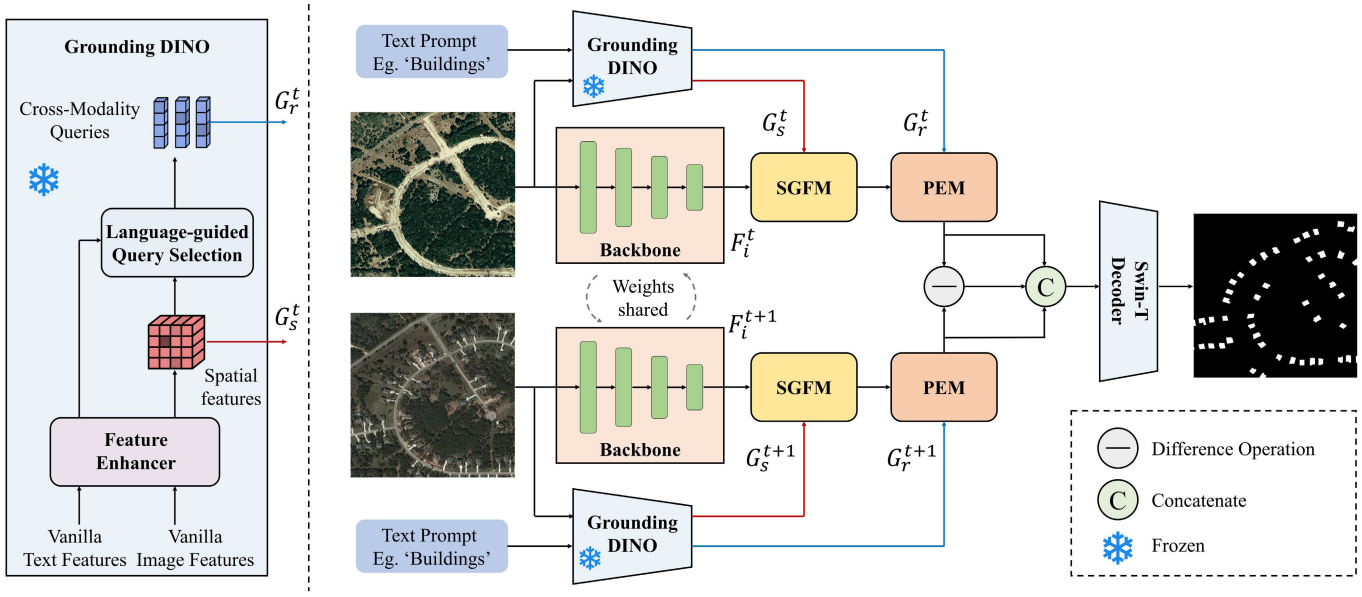



Fig. 2. Overall architecture of ChangeDINO.

address the two structural challenges. Specifically, the SGFM addresses the spatially heterogeneous activation distribution inherent to open-vocabulary detector features by adaptively exploiting high-activation semantic regions as guidance signals while suppressing uninformative background responses. The PEM addresses the granularity gap between object-level positional priors and pixel-level spatial guidance by transforming compact positional representations into per-pixel spatial modulation signals through feature reorganization and parametric modulation. Finally, a multilevel decoder integrates bi-temporal information, reconstructing features from high to low levels to achieve accurate change detection.

### B. Semantically Guided Fusion Module

*1) Design Rationale and Mechanism:* The SGFM is designed to leverage semantic priors from Grounding DINO's feature enhancer, which produces spatially aligned features with rich semantic information through multilayer cross-attention between visual and language modalities. The key insight is that semantic features have varying reliability across spatial locations—some regions contain certain semantic information while others may be ambiguous.

To achieve high-quality feature guidance, we propose SGFM. As shown in Fig. 3, for input at time $t$, we first preprocess both visual and semantic features to ensure consistency in feature representation. For visual features $F^t_{v,i}$, feature projection is applied

$$F^t_{p,i} = \mathrm{ReLU}\left(\mathrm{GN}\left(\mathrm{Conv1}\times 1\left(F^t_{v,i}\right)\right)\right) \tag{1}$$

where $F^t_{p,i} \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$ represents the projected visual features, and GN denotes group normalization. For semantic features, we use convolution and bilinear interpolation to align their spatial dimensions with visual features $G^t_{s,i} \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$. At the macro level, we employ information entropy [47] to quantify feature certainty, providing reliability metrics for subsequent feature guidance. To transform

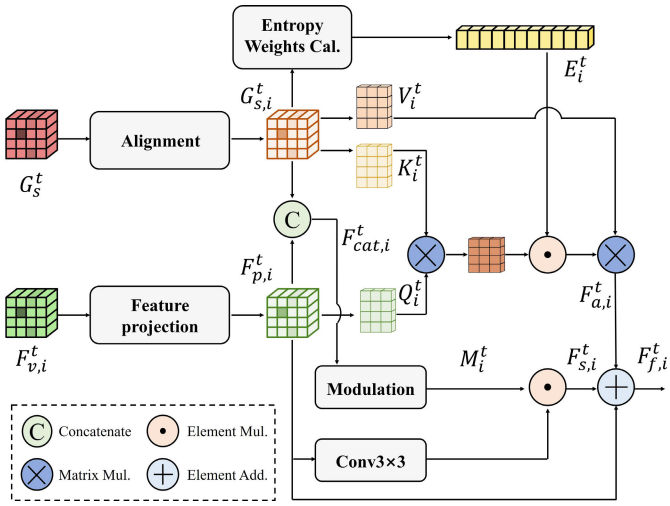



Fig. 3. Structure of the SGFM.

pixel distributions into probability distributions while ensuring numerical stability, we apply the Softmax to semantic features $G^t_{s,i} \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$ at layer $i$. Specifically, we construct the entropy evaluation as follows:

$$p^t_{b,c,h,w,i} = \frac{\exp\left(g^t_{b,c,h,w,i}\right)}{\sum_{h',w'}\exp\left(g^t_{b,c,h',w',i}\right)} \tag{2}$$

$$H\left(G^t_{s,i}\right)_{b,c} = -\sum_{h,w} p^t_{b,c,h,w,i}\log_2\left(p^t_{b,c,h,w,i}+\epsilon\right) \tag{3}$$

where $g^t_{b,c,h,w,i}$ represents the value at spatial position $(h,w)$ in channel $c$ of batch $b$ in the semantic feature map, $p^t_{b,c,h,w,i}$ denotes the normalized probability distribution, and $\epsilon = 1\mathrm{e}^{-8}$ is a numerical stability constant. By computing entropy independently for each channel, we compress spatial information $H_i \times W_i$ into a single entropy value, obtaining $H(G^i_s) \in \mathbb{R}^{B\times C}$.

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Different entropy values indicate varying degrees of semantic information richness, with lower entropy values corresponding to more prominent semantic content. Based on this, we design an inverse entropy weighting strategy

$$E_i^t = 1 - \frac{H\left(G_{s,i}^t\right) - \min\left(H\left(G_{s,i}^t\right)\right)}{\max\left(H\left(G_{s,i}^t\right)\right) - \min\left(H\left(G_{s,i}^t\right)\right)}. \tag{4}$$

In designing the feature guidance mechanism, establishing effective associations between visual and semantic features is crucial. Visual features typically contain rich spatial details and local structural information but lack semantic understanding, while semantic features are more accurate in object-level representation but may lose detailed information. Based on this observation, we designate visual features as query and semantic features as key and value, enabling visual features to actively "query" relevant information from semantic features, thereby achieving knowledge transfer from semantic to visual domains. Specifically

$$\begin{aligned} Q_i^t &= W_{q,i}^t\left(F_{p,i}^t\right) \\ K_i^t &= W_{q,i}^t\left(G_{s,i}^t\right), \quad V_i^t = W_{v,i}^t\left(G_{s,i}^t\right) \end{aligned} \tag{5}$$

where $W_{q,i}^t, W_{k,i}^t, W_{v,i}^t \in \mathbb{R}^{C_i \times C_i}$ are learnable linear transformation matrices for generating query, key, and value features. When computing attention weights, we introduce an entropy-guided mechanism

$$F_{a,i}^t = \text{Softmax}\left(\frac{Q_i^t K_i^{t^T}}{\sqrt{d}} \odot E_i^t\right) V_i^t. \tag{6}$$

Furthermore, we concatenate visual and semantic features along the channel dimension to capture correlations between the two types of features

$$F_{cat,i}^t = \left[F_{p,i}^t; G_{s,i}^t\right] \in \mathbb{R}^{B \times 2C_i \times H_i \times W_i}. \tag{7}$$

Through two linear transformations and nonlinear activation, we generate spatially adaptive modulation maps

$$M_i^t = \sigma\left(W_{2,i}^t \text{ReLU}\left(W_{1,i}^t\left(F_{cat,i}^t\right)\right)\right) \tag{8}$$

where $W_{1,i}^t \in \mathbb{R}^{C_i \times 2C_i}$ and $W_{2,i}^t \in \mathbb{R}^{1 \times C_i}$ are weight matrices for the two linear transformations, and $\sigma$ represents the Sigmoid function. This modulation map $M_i^t \in \mathbb{R}^{B \times 1 \times H_i \times W_i}$ encodes the degree to which visual features at each spatial position should be enhanced or suppressed. Finally, we multiply the modulation map with convolution-enhanced visual features to achieve adaptive feature modulation

$$F_{s,i}^t = \text{Conv3} \times 3\left(F_{p,i}^t\right) \odot M_i^t. \tag{9}$$

The final feature generation is achieved through a residual connection for information fusion

$$F_{f,i}^t = F_{p,i}^t + F_{a,i}^t + F_{s,i}^t. \tag{10}$$

SGFM achieves high-quality feature guidance through three mechanisms.

1) *Entropy-Based Reliability Assessment [see (3) and (4)]:* Information entropy quantifies the degree of semantic certainty across spatial locations, with different entropy values reflecting varying levels of semantic concentration. The inverse weighting $E_i^t$ adaptively modulates channel contributions according to their semantic certainty.
2) *Semantic-Guided Attention [see (5) and (6)]:* Visual features serve as query to actively "query" relevant information from semantic features (key and value), modulated by entropy weights to selectively attend to reliable semantic regions.
3) *Spatial-Adaptive Modulation [see (8) and (9)]:* The modulation map $M_i^t$ performs pixel-wise calibration, enhancing semantically relevant regions while suppressing false alarms.

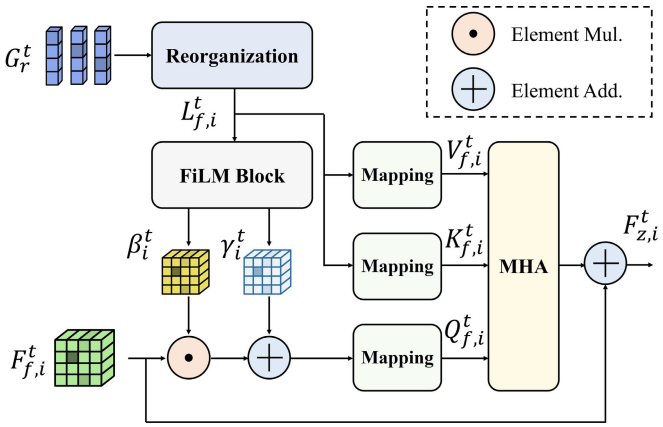



Fig. 4. Structure of the PEM.

### C. Position-Aware Enhancement Module

*1) Design Rationale and Mechanism:* The PEM extracts positional features from Grounding DINO's interactive decoder outputs $G_r^t$, which encode not only object locations but also spatial relationships through language-guided query selection and multiscale feature interaction. Unlike simple coordinate embeddings, these features capture contextual spatial information.

Positional information from object detection models holds significant value for change detection. To fully leverage this advantage, we extract positional features from the Grounding DINO decoder, which not only contains object location information but also preserves spatial relationship representations. Specifically, the positional features $G_r^t$, $G_r^{t+1} \in \mathbb{R}^{N_d \times D}$ are obtained from the interactive decoder, where $N_d$ denotes the number of decoder queries, which is derived from the image high-level features through reshaping, concatenation, and language-guided processing. These positional features encode precise location information of targets and spatial relationships between targets and their surrounding environment.

As shown in Fig. 4, given features $G_r^t$ extracted from the Grounding DINO decoder at time $t$, we first achieve spatial reconstruction of positional features through a feature reorganization network

$$L_{s,i}^t = W_{4,i}^t\left(\text{Reshape}\left(W_{3,i}^t\left(G_r^t\right)\right)\right) \tag{11}$$

where $G_r^t \in \mathbb{R}^{B \times N \times D}$ represents decoder features containing rich positional information, $L_{s,i}^t \in \mathbb{R}^{B \times D \times H_i \times W_i}$ denotes the reorganized spatial positional features, and $W_{3,i}^t \in \mathbb{R}^{D \times (H_i W_i)}$

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and $W^t_{4,i} \in \mathbb{R}^{D\times D}$ are learnable transformation matrices. This reconstruction design aligns positional and visual features in spatial dimensions, establishing a foundation for subsequent feature interaction.

After positional feature reconstruction, we employ the FiLM mechanism to achieve feature modulation. Modulation parameters are generated through a nonlinear transformation network

$$\left[\gamma^t_i, \beta^t_i\right] = W^t_{6,i}\left(\mathrm{ReLU}\left(W^t_{5,i}\left(L^t_{s,i}\right)\right)\right) \tag{12}$$

where the generated parameters $\gamma^t_i, \beta^t_i \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$ serve as multiplicative and additive factors, respectively, and $W^t_{5,i} \in \mathbb{R}^{D\times C_i}$ and $W^t_{6,i} \in \mathbb{R}^{C_i\times 2C_i}$ are learnable transformation matrices. This parameterized modulation approach enables the model to dynamically adjust feature responses based on positional information, achieving precise feature calibration through pixel-level modulation

$$F^t_{m,i} = \gamma^t_i \odot F^t_{f,i} + \beta^t_i \tag{13}$$

$$\left[Q^t_{f,i}, K^t_{f,i}, V^t_{f,i}\right] = \left[W^i_q\left(F^t_{m,i}\right), W^i_k\left(L^t_{s,i}\right), W^i_v\left(L^t_{s,i}\right)\right] \tag{14}$$

where $W^i_q, W^i_k, W^i_v \in \mathbb{R}^{D\times d}$ are projection matrices, $d$ represents the attention head dimension, and $Q^t_{f,i}, K^t_{f,i}, V^t_{f,i} \in \mathbb{R}^{B\times(H_iW_i)\times d}$. Through parallel multihead attention computation, each spatial position can adaptively aggregate relevant positional information, achieving more precise feature enhancement. The final feature

$$F^t_{z,i} = \mathrm{MHA}\left(Q^t_{f,i}, K^t_{f,i}, V^t_{f,i}\right) + F^t_{f,i}. \tag{15}$$

### D. Decoder and Loss Function

*1) Decoder:* After feature enhancement, a Swin Transformer [48] decoder is employed to perform change detection. First, we calculate the difference between bi-temporal features

$$F_{diff,i} = \left\| F^t_{z,i} - F^{t+1}_{z,i} \right\|. \tag{16}$$

Then, it fuses the bi-temporal features. The decoding process gradually refines features in a cascaded manner from high to low levels.

*2) Loss Function:* We use the cross-entropy loss function [49]

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\hat{y}_i\right) + \left(1-y_i\right)\log\left(1-\hat{y}_i\right)\right] \tag{17}$$

where $N$ represents the total number of pixels, $y_i$ denotes the ground truth label of the $i$th pixel, and $\hat{y}_i$ represents the predicted probability of the $i$th pixel by the model.

PEM processes positional features through two stages.

1) *Spatial Reorganization [see (11)]:* Decoder features $G^t_r$ in compact query format ($N_d \times D$) are transformed into spatial format ($D \times H_i \times W_i$) via learnable projections, aligning with visual feature dimensions while preserving positional encoding.
2) *FiLM-Based Calibration [see (12) and (13)]:* The reorganized features $L^t_{s,i}$ generate modulation parameters $\gamma^t_i$ (scale) and $\beta^t_i$ (shift) through nonlinear transformation, performing pixel-wise feature adjustment based on positional priors.

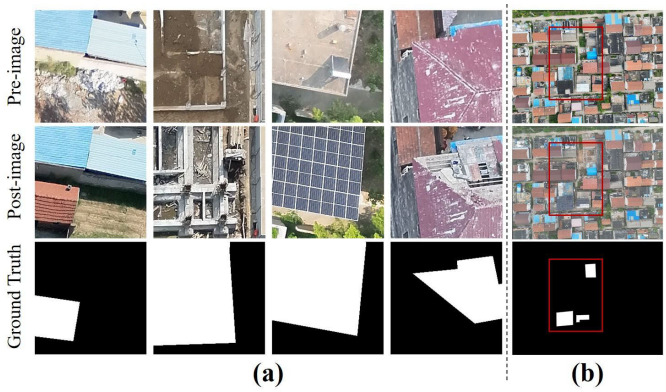



Fig. 5. Examples from the AUUCDD dataset. (a) Four typical unauthorized construction cases after patch segmentation (256 × 256 pixels): facade alteration (first column), unauthorized expansion (second column), illegal solar panel installation (third column), and structural modification (fourth column). (b) Example of original high-resolution image (3024 × 4032 pixels).

TABLE I
COMPARISON OF AUUCDD WITH EXISTING UAV-BASED CHANGE DETECTION DATASETS

| Dataset | Original Image Size | Total Image Pairs | Resolution | Change Categories |
|---|---|---|---|---|
| UAV-BCD | 768×768 | 2,024 | 0.06 m | Building additions, demolitions |
| UAV-CD | 768×768 | 2,660 | 0.06 m | Building changes; land excavation |
| UAV-CD+ | 1024×1024 | 2,002 | 0.06 m | Urban architecture, construction site, land, water body changes |
| **AUUCDD** | **3024×4032** | **2,000** | **0.06 m** | **Facade alteration, unauthorized expansion, illegal solar panel installation, structural modification** |

## IV. EXPERIMENT

### A. Datasets and Evaluation Metrics

We conduct experimental validation on four public benchmark remote sensing datasets—LEVIR-CD [50], WHU-CD [34], SYSU-CD [12], and SECOND [51]—as well as our collected AUUCDD drone-based dataset.

**LEVIR-CD** contains 637 pairs of high-resolution (1024 × 1024 pixels) bi-temporal remote sensing images. Following the official split of LEVIR-CD, we use 445 pairs for training, 64 pairs for validation, and 128 pairs for testing. For our experiments, each image pair is split into nonoverlapping patches of 256 × 256 pixels.

**WHU-CD** consists of two large-scale aerial images (32 507 × 15 354 pixels) captured at different times. After dividing into nonoverlapping 256 × 256 patches, we obtain 5948/744/744 image pairs for training, validation, and testing, respectively [34].

**SYSU-CD** is a large-scale seasonal change detection dataset containing 20 000 pairs of aerial images (256 × 256 pixels) captured across different seasons in Hong Kong. The dataset focuses on urban building changes and provides pixel-level annotations. Following the official split, we use 12 000/4000/4000 image pairs for training, validation, and testing, respectively. This dataset is particularly challenging due to seasonal variations in lighting, vegetation, and weather conditions.

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**AUUCDD** focuses on urban unauthorized construction monitoring. It is collected through a grid-based UAV monitoring platform using DJI Matrice3TD drones along more than 1000 fixed flight routes at 120 m over Xuzhou City, China, capturing representative urban scenes typical of routine monitoring tasks. Fig. 5 shows some examples of AUUCDD. Prior to dataset creation, we performed pixel-level registration between bi-temporal images to ensure precise alignment [52]. The dataset comprises 2000 high-resolution images ($3024 \times 4032$ pixels), primarily documenting unauthorized building modifications and illegal solar panel installations, with all data labeled under the guidance of domain experts. Each image is segmented into nonoverlapping patches of $256 \times 256$ pixels. And 12 000/3000/3000 image pairs are generated for training, validation, and testing, respectively. To contextualize AUUCDD within the existing UAV-based change detection landscape, Table I compares it with three representative datasets: UAV-BCD [53], UAV-CD [54], and UAV-CD+ [55], which cover general surface changes including building, land, and water area variations. Compared to existing UAV-based datasets, AUUCDD stands out with its significantly larger image dimensions ($3024 \times 4032$ pixels), which preserves richer spatial detail for detecting fine-grained unauthorized construction activities, making it more representative of real-world low-altitude UAV monitoring deployments.

**SECOND** is a large-scale seasonal change detection dataset containing 4662 pairs of high-resolution bi-temporal remote sensing images. The dataset provides multiclass semantic annotations covering six land cover change types: bare soil (including sand), trees, low vegetation (grassland, shrubs, etc.), water, buildings, and playgrounds. The diverse land cover types in the SECOND dataset make it an ideal choice for validating the generalization capability of change detection methods.

*1) Evaluation Metrics:* To quantitatively evaluate the performance of our proposed method, we adopt three widely used metrics in change detection tasks: $F1$-score, intersection over union (IoU), and recall. These metrics are computed as follows:

$$Precision = \frac{TP}{TP + FP} \tag{18}$$

$$Recall = \frac{TP}{TP + FN} \tag{19}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{20}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{21}$$

where TP represents the number of correctly detected change pixels, FP represents the number of unchanged pixels incorrectly classified as changed, and FN represents the number of changed pixels incorrectly classified as unchanged. The $F1$-score provides a balanced measure of both precision and recall, IoU measures the overlap between predicted and ground truth change regions, and recall indicates the model's ability to detect all actual changes.

### B. Experiment Settings

For model optimization, we use the AdamW optimizer with a learning rate of $1e^{-4}$ and weight decay of 0.01. All experiments are conducted on a single NVIDIA Tesla A100 GPU with a batch size of 16. The model is trained for 40 K iterations in both comparative experiments and ablation studies, while for data-scarce experiments, we train for 20 K iterations. For data augmentation, we employ RandomFlip and PhotoMetricDistortion [10].

For Grounding DINO integration, we employ the Swin-T-OGC version. We use different prompts for different datasets: a simple prompt "buildings" for LEVIR-CD, SYSU-CD, and WHU-CD, while for the AUUCDD, we use "building modifications. steel structures" to better capture unauthorized construction characteristics.

### C. Comparison With SOTA

We conduct comprehensive comparisons between our proposed ChangeDINO method and several SOTA change detection approaches, including: STNet [56], SARASNet [57], AFCF3DNet [58], DDLNet [59], ChangeCLIP [29], MMChange [46], HCGMNet [60], ChangeMamba [39], CGNet [61], C2FNet [62], SNUNet-CD [18], B2CNet [63], LSKNet [64], and ScratchFormer [65].

*1) Quantitative Performance Analysis:* As shown in Table II, ChangeDINO achieves the best $F1$-scores across all four datasets, and the best IoU on LEVIR-CD, SYSU-CD, and AUUCDD. On the LEVIR-CD dataset, using ResNet50 as the backbone network, ChangeDINO attains an $F1$-score of 92.89% and IoU of 86.17%, representing improvements of 0.62% and 0.52% over LSKNet, respectively. On the WHU-CD dataset, our method achieves an $F1$-score of 95.79%, surpassing all compared methods. The IoU of 90.01% is slightly below MMChange's 90.90%, which is attributed to the different design emphases of the two methods: MMChange incorporates a dedicated image feature refinement module to impose fine-grained boundary constraints on change regions, offering a unique advantage in boundary precision; whereas ChangeDINO focuses more on complete detection of change regions under semantic guidance, leading to superior performance on the F1 metric. On the SYSU-CD dataset, which presents additional challenges due to seasonal variations, ChangeDINO achieves an $F1$-score of 83.87%, outperforming ChangeCLIP and ScratchFormer by 1.15% and 2.18% in F1, respectively. On our newly proposed AUUCDD dataset, ChangeDINO attains an $F1$-score of 90.17% and IoU of 82.34%, achieving superior detection results compared to other methods, demonstrating the effectiveness of our approach in handling low-altitude drone imagery and unauthorized construction change detection tasks. We also compare ChangeDINO with two efficiency-oriented baselines, SNUNet-CD and B2CNet. Despite their lower computational overhead, both methods consistently underperform ChangeDINO across all datasets, confirming that the additional complexity introduced by Grounding DINO yields meaningful and irreplaceable performance gains.

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TABLE II
QUANTITATIVE COMPARISON WITH SOTA ON FOUR DATASETS (%)

| Method | Backbone | LEVIR-CD | | | WHU-CD | | | SYSU-CD | | | AUUCDD | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | F1↑ | IoU↑ | Rec↑ | F1↑ | IoU↑ | Rec↑ | F1↑ | IoU↑ | Rec↑ | F1↑ | IoU↑ | Rec↑ |
| STNet | ResNet18 | 90.25 | 82.02 | 89.30 | 87.64 | 77.72 | 87.08 | 79.82 | 66.42 | 77.25 | 83.42 | 73.12 | 79.18 |
| SARASNet | ResNet18 | 90.40 | 82.49 | 89.95 | 90.48 | 82.62 | 89.58 | 80.35 | 67.15 | 78.03 | 83.76 | 75.45 | 77.42 |
| AFCF3DNet | ResNet18 | 90.76 | 83.08 | 90.17 | 93.58 | 87.93 | 93.69 | 80.57 | 67.71 | 74.61 | 86.87 | 79.08 | 85.42 |
| DDLNet | ResNet18 | 90.60 | 82.49 | 89.41 | 90.56 | 82.75 | 90.03 | 79.24 | 65.58 | 76.89 | 81.43 | 77.54 | 83.19 |
| SNUNet-CD | – | 89.23 | 81.87 | 88.34 | 88.76 | 83.43 | 89.12 | 78.89 | 66.74 | 77.23 | 83.67 | 75.84 | 82.91 |
| B2CNet | ResNet18 | 89.56 | 82.13 | 88.79 | 89.34 | 83.87 | 89.67 | 79.34 | 67.12 | 77.68 | 84.23 | 76.31 | 83.45 |
| ChangeCLIP | ResNet18 | 89.04 | 83.47 | 90.21 | 92.78 | 89.31 | 92.49 | 81.72 | 70.03 | 80.41 | 87.71 | 79.34 | 85.08 |
| MMChange | ResNet18 | 91.57 | 84.44 | 90.78 | 94.90 | 90.30 | 93.36 | 83.64 | 71.88 | 81.42 | 88.01 | 79.21 | 85.46 |
| ChangeDINO | ResNet18 | 91.57 | 84.99 | 91.49 | 93.89 | 89.57 | 93.66 | 82.01 | 70.68 | 80.15 | 88.23 | 80.18 | 87.86 |
| HCGMNet | VGG16 | 91.77 | 84.79 | 90.61 | 92.08 | 85.33 | 90.31 | 79.76 | 66.33 | 74.15 | 83.43 | 74.52 | 82.31 |
| Changemamba | VMamba-S | 91.99 | 83.94 | 91.25 | 94.19 | 89.02 | 92.23 | 82.46 | 70.59 | 80.01 | 86.64 | 78.86 | 84.19 |
| CGNet | VGG16 | 92.01 | 85.21 | 90.90 | 92.59 | 86.21 | 90.79 | 79.92 | 66.45 | 74.37 | 85.64 | 76.83 | 80.42 |
| C2FNet | VGG16 | 90.83 | 84.89 | 91.04 | 94.36 | 89.83 | 92.76 | 81.45 | 68.76 | 78.92 | 86.76 | 77.19 | 82.41 |
| ScratchFormer | – | 91.68 | 84.99 | 90.34 | 95.16 | 89.20 | 92.43 | 81.69 | 69.26 | 78.62 | 85.63 | 78.08 | 81.29 |
| ChangeCLIP | ResNet50 | 92.01 | 85.20 | 90.67 | 94.82 | 90.15 | 92.02 | 82.72 | 70.53 | 80.41 | 87.98 | 80.54 | 85.84 |
| ChangeCLIP | ViT-B | 91.30 | 83.99 | 89.04 | 94.78 | 90.08 | 92.78 | 83.32 | 71.41 | 79.80 | 87.65 | 79.39 | 84.98 |
| LSKNet | LSK-S | 92.27 | 85.65 | 91.23 | 93.37 | 89.04 | 91.45 | 81.88 | 69.35 | 79.56 | 85.19 | 75.72 | 81.17 |
| MMChange | ResNet50 | 91.93 | 85.06 | 91.06 | 95.23 | 90.90 | 93.68 | 83.76 | 72.05 | 81.04 | 89.01 | 81.47 | 88.35 |
| ChangeDINO | ResNet50 | 92.89 | 86.17 | 91.88 | 95.79 | 90.01 | 94.79 | 83.87 | 71.86 | 80.76 | 90.17 | 82.34 | 89.13 |
| ChangeDINO | ViT-B | 92.46 | 85.64 | 91.73 | 95.01 | 89.31 | 93.46 | 83.46 | 71.44 | 80.45 | 89.58 | 81.76 | 88.67 |

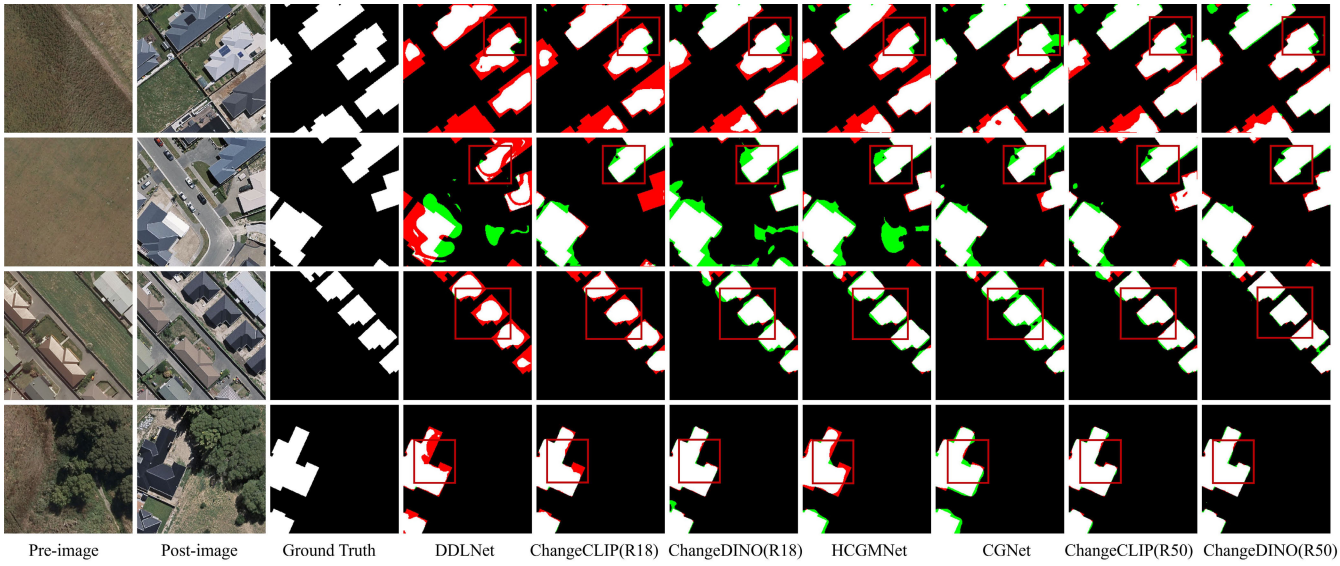



Fig. 6. Qualitative comparison with SOTA on the WHU-CD. Red regions indicate false negatives, and green regions indicate false positives.

Both MMChange and ChangeDINO leverage prior knowledge from large-scale vision–language pretrained models to enhance change detection performance, and both achieve highly competitive results across multiple datasets, jointly validating the effectiveness of vision–language models in RSCD. The distinction lies in their design philosophy: MMChange introduces semantic modeling at the text prompt level, while ChangeDINO directly exploits the internal outputs of an open-vocabulary detector at the feature level through SGFM and PEM. The two methods explore the potential of vision–language priors from different perspectives, and both achieve satisfying detection results. Compared to ChangeCLIP, which also leverages vision–language models, ChangeDINO achieves superior results on most metrics, primarily owing to our bi-level feature entanglement design, which effectively integrates the semantic and positional information of the open-vocabulary detector into pixel-level change detection through SGFM and PEM, enabling more fine-grained semantic guidance and spatial localization.

*2) Qualitative Results Analysis:* Figs. 6–9 present qualitative comparison results across the four datasets. ChangeDINO demonstrates superior performance in several key aspects.

*Boundary Precision:* Compared to other methods, ChangeDINO achieves more accurate localization of change region boundaries. In the WHU-CD results shown in Fig. 6, our method significantly outperforms DDLNet, ChangeCLIP,

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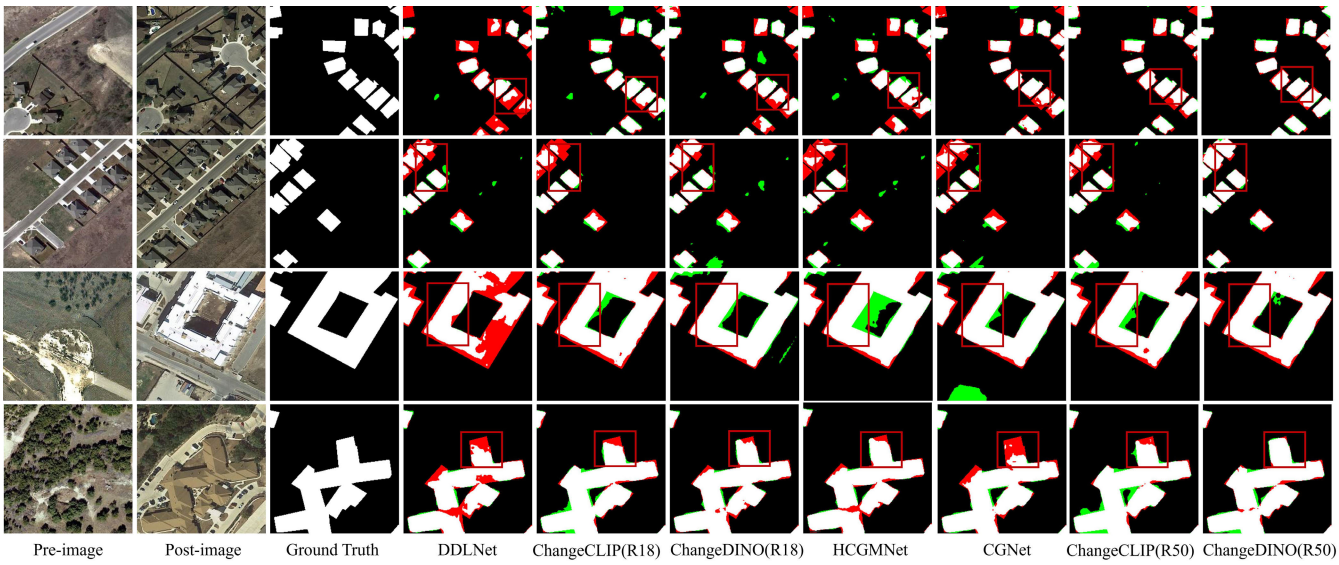



Fig. 7. Qualitative comparison with SOTA on the LEVIR-CD. Red regions indicate false negatives, and green regions indicate false positives.

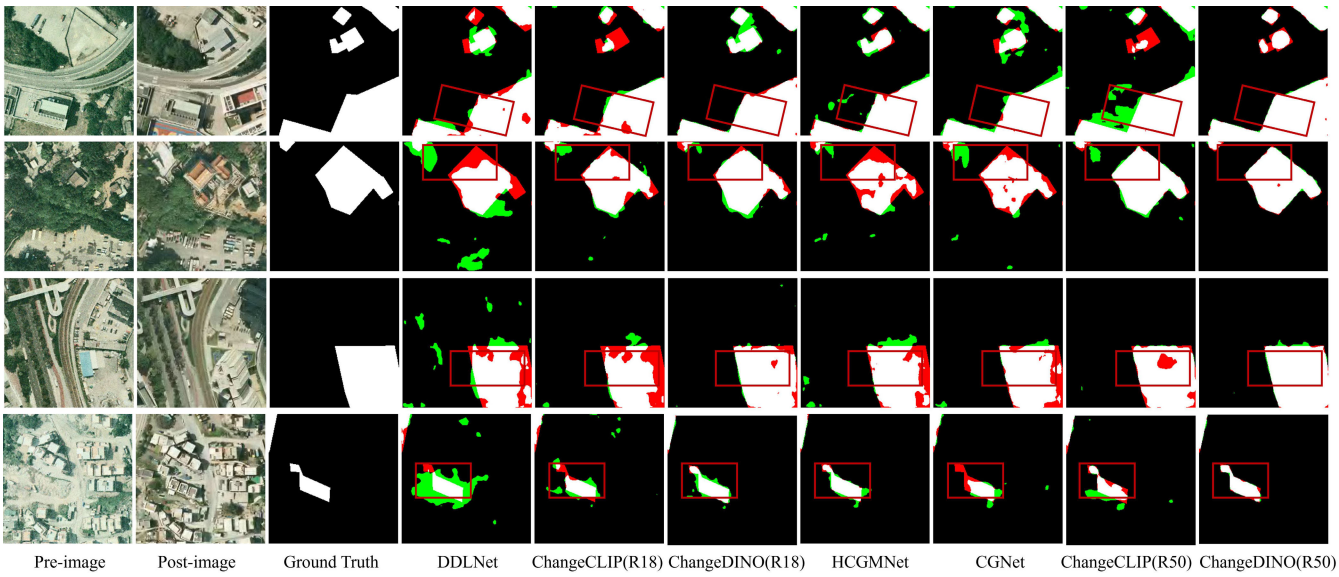



Fig. 8. Qualitative comparison with SOTA on the SYSU-CD. Red regions indicate false negatives, and green regions indicate false positives.

and HCGMNet in building edge detection, reducing boundary blurring and incomplete detection issues.

*Detection Completeness:* In the LEVIR-CD results shown in Fig. 7, ChangeDINO can more completely detect change regions, particularly when handling large building changes. Other methods, such as DDLNet and CGNet, frequently exhibit incomplete detection (manifested as red regions indicating false negatives), while our method maintains detection completeness.

*False Alarm Suppression:* ChangeDINO excels in suppressing false alarms. Across all test scenarios, our method produces significantly fewer green regions (false positives) compared to other comparison methods, indicating that the semantic guidance mechanism effectively improves detection accuracy.

*Seasonal Robustness:* In the SYSU-CD results shown in Fig. 8, ChangeDINO demonstrates strong robustness to seasonal variations. Despite challenges posed by lighting changes and seasonal differences in vegetation appearance, our method maintains consistent detection performance with fewer false alarms compared to baseline methods.

*Complex Scene Adaptability:* In the AUUCDD results shown in Fig. 9, when facing complex scenarios in drone aerial imagery, including varying lighting conditions, shadow changes, and complex backgrounds, ChangeDINO maintains stable detection performance, while other methods often exhibit more misdetections.

*3) Performance Advantage Analysis:* Compared to traditional CNN-based methods, ChangeDINO achieves significant improvements in semantic understanding capability by incorporating prior knowledge from pretrained vision–language models. Unlike methods that primarily rely on training data, our approach leverages the rich semantic and positional information from Grounding DINO, enabling

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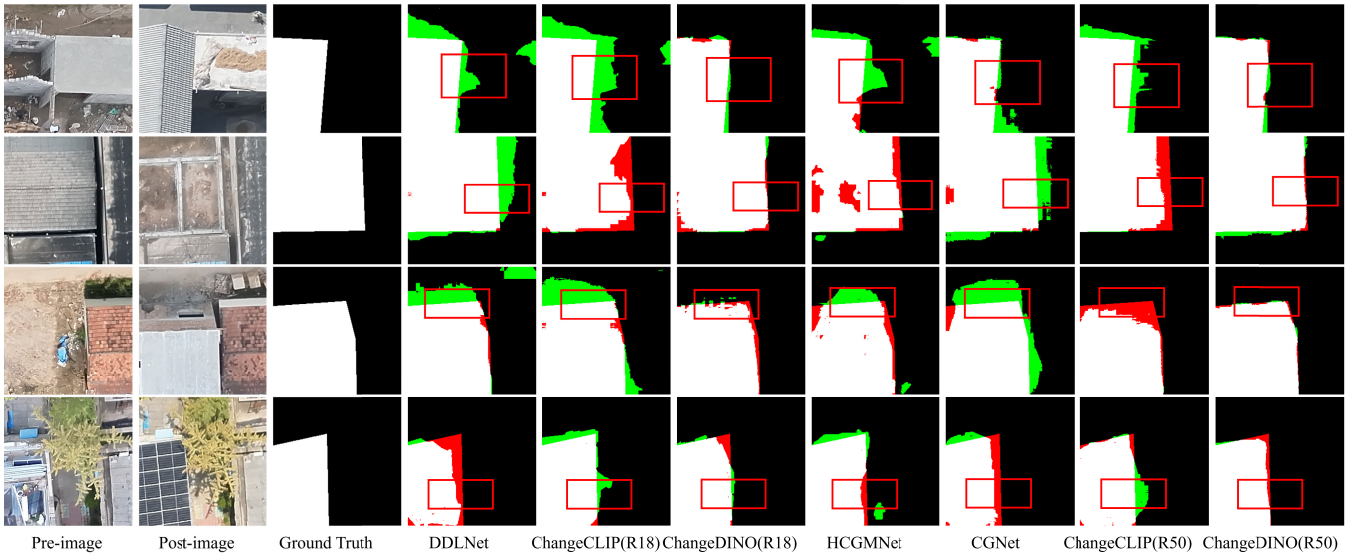



Fig. 9. Qualitative comparison with SOTA on the AUUCDD. Red regions indicate false negatives, and green regions indicate false positives.

TABLE III
ABLATION STUDIES ON PEM AND SGFM MODULES WITH RESNET50 AND VIT-B (%)

| Backbone | Method | PEM | SGFM | LEVIR-CD | | | AUUCDD | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | F1↑ | IoU↑ | Rec↑ | F1↑ | IoU↑ | Rec↑ |
| ResNet50 | Baseline | | | 88.78 | 82.27 | 88.25 | 85.43 | 77.31 | 84.19 |
| | w/PEM | ✓ | | 90.21 | 83.94 | 89.42 | 87.21 | 79.09 | 85.86 |
| | w/SGFM | | ✓ | 91.32 | 84.66 | 89.82 | 87.64 | 79.31 | 86.08 |
| | Full Model | ✓ | ✓ | **92.89** | **86.17** | **91.88** | **90.17** | **82.34** | **89.13** |
| ViT-B | Baseline | | | 89.47 | 81.94 | 88.27 | 84.31 | 77.53 | 83.64 |
| | w/PEM | ✓ | | 90.89 | 83.78 | 89.46 | 86.64 | 78.86 | 85.86 |
| | w/SGFM | | ✓ | 91.36 | 84.05 | 89.90 | 86.95 | 80.09 | 86.63 |
| | Full Model | ✓ | ✓ | **92.46** | **85.64** | **91.73** | **89.58** | **81.76** | **88.67** |

the model to better understand the semantic meaning of changes.

Compared to ChangeCLIP, although both utilize vision–language models, the key difference lies in the granularity of information. CLIP can only provide global scene-level semantic information, while Grounding DINO provides object-level precise localization and richer positional information. This difference is particularly evident in recall metrics: our method achieves recall rates of 91.88%, 94.79%, and 89.13% on LEVIR-CD, WHU-CD, and AUUCDD, respectively, all superior to other comparison methods.

Experimental results demonstrate that ChangeDINO successfully integrates semantic knowledge and positional information through the bi-level feature entanglement mechanism, improving detection completeness while maintaining high precision, providing an effective solution for RSCD tasks.

### D. Ablation Study

*1) Ablation Analysis of Overall Architecture:* To comprehensively understand the bi-level feature entanglement mechanism in ChangeDINO, we design systematic ablation experiments. Specifically, we construct four model variants to validate the contributions of the SGFM module and the PEM module. The baseline model only contains dual-stream feature extraction and simple feature fusion. To isolate the effect of each module, we further introduce the PEM and SGFM separately, resulting in the w/PEM and w/SGFM variants.

The experimental results (as shown in Table III) reveal the crucial roles of semantic guidance and position awareness in change detection. With ResNet50 as the backbone, incorporating only the PEM module improves the $F1$-score on LEVIR-CD from 88.78% to 90.21%, attributable to enhanced modeling of spatial relationships through position-aware features. In contrast, incorporating only the SGFM module achieves a 91.32% $F1$-score, owing to the introduction of semantic information. Notably, the full model with both PEM and SGFM further improves performance to 92.89%, validating the significance of the bi-level feature entanglement mechanism in enhancing feature discriminability.

Fig. 10 presents the feature map visualizations that provide intuitive insights into how different module combinations affect feature representation. The visualized features correspond to the first-layer features ($64 \times 64$ resolution) under different configurations: the baseline model shows the projected basic visual features $F^t_{p,1}$, the w/PEM version displays features processed through the PEM $F^t_{z,1}$, the w/SGFM version presents features from the SGFM $F^t_{f,1}$, and the full model

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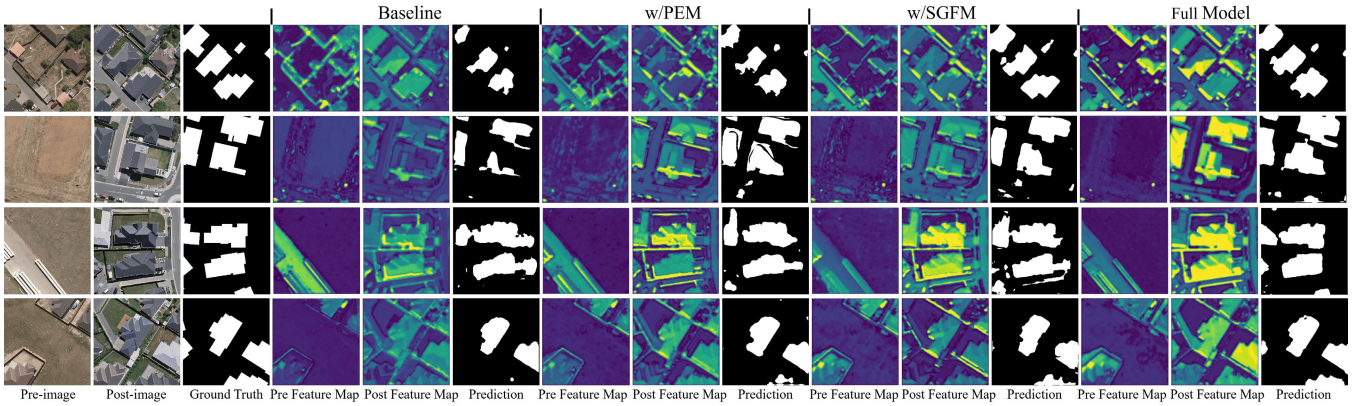



Fig. 10. Visualization of prediction results and feature maps with different module combinations on the WHU-CD.

exhibits the final features $F^t_{z,1}$ after sequential processing through both modules.

The baseline feature maps exhibit relatively scattered activation patterns with weak responses to change-relevant regions. The feature activations lack spatial coherence and appear diffuse across the feature space, indicating limited discriminative capability for change detection tasks.

With the introduction of the PEM module (w/PEM), the feature maps demonstrate enhanced spatial structure and more concentrated activations. The position-aware mechanism effectively reorganizes the feature responses, leading to clearer spatial boundaries and more focused activation patterns around potential change areas.

The SGFM module (w/SGFM) produces feature maps with notably improved semantic selectivity. The activations show stronger responses to semantically relevant regions while suppressing background noise. The entropy-guided attention mechanism successfully enhances the discriminative power of features for building-related changes.

The full model combining both modules generates the most refined feature representations. The feature maps exhibit both spatial coherence from position awareness and semantic selectivity from guided fusion. The activation patterns are well-organized, concentrated, and semantically meaningful, demonstrating the synergistic effect of the bi-level feature entanglement mechanism.

To investigate the effectiveness of each component in the SGFM, we design a series of ablation variants as follows:

This progressive enhancement in feature quality clearly illustrates how each proposed module contributes to improved feature learning, providing strong visual evidence for the effectiveness of our architectural design.

*2) Analysis of SGFM:*

1) *Baseline:* Basic model: $F^t_{f,i} = F^t_{p,i}$.
2) *w/ Feature Concat:* Incorporates semantic features through concatenation: $F^t_{f,i} = F^t_{p,i} + \text{Conv}([F^t_{p,i}; G^t_{s,i}])$.
3) *w/ Semantic Guide:* Employs semantic-guided attention: $F^t_{f,i} = F^t_{p,i} + F^t_{a,i}$, where $F^t_{a,i}$ not modulated by entropy $E^t_i$.
4) *w/ Entropy Weight:* Adds feature modulation: $F^t_{f,i} = F^t_{p,i} + F^t_{a,i}$.
5) *Full Model:* Complete model: $F^t_{f,i} = F^t_{p,i} + F^t_{a,i} + F^t_{s,i}$.

We conduct ablation studies of SGFM on both LEVIR-CD and AUUCDD, as shown in Table IV. The baseline model, which only contains basic dual-stream feature extraction, achieves $F1$-scores of 90.21% and 87.21% on LEVIR-CD and AUUCDD datasets, respectively. Building upon this foundation, we introduce a feature fusion strategy (w/ Feature Concat) that combines visual and semantic features through concatenation followed by convolution operations. Experimental results demonstrate that this simple feature fusion strategy yields modest performance improvements, with $F1$-scores increasing to 90.56% on LEVIR-CD and 87.53% on AUUCDD.

By incorporating the semantic guidance mechanism (w/ semantic guide) that establishes attention-based interactions between visual and semantic features, enabling more effective utilization of semantic information. This enhancement significantly improves model performance, achieving $F1$-scores of 91.32% and 88.19% on the respective datasets, demonstrating the superiority of active semantic guidance over simple feature concatenation. Notably, the introduction of the entropy weighting mechanism (w/ entropy weight) further improves model performance, achieving $F1$-scores of 92.25% on LEVIR-CD and 89.31% on AUUCDD.

To explore the efficacy of each component within the PEM, we design multiple ablation variants as follows:

*3) Analysis of PEM:*

1) *Baseline:* $F^t_{z,i} = F^t_{f,i}$.
2) *w/* $L^t_{s,i}$: $F^t_{z,i} = F^t_{f,i} + \text{Conv}([F^t_{f,i}, L^t_{s,i}])$.
3) *w/ Attention:* $F^t_{z,i} = F^t_{f,i} + \text{Attn}(F^t_{f,i}, L^t_{s,i})$, where $F^t_{f,i}$ without FiLM modulation.
4) *w/ Full Model:* $F^t_{z,i} = F^t_{f,i} + F^t_{l,i}$.

As shown in Table V, the baseline model achieves $F1$-scores of 91.32% and 87.64% on LEVIR-CD and AUUCDD datasets, respectively. By incorporating positional features from Grounding DINO through direct concatenation (w/ $L^t_{s,i}$), the model exhibits initial performance improvements, with $F1$-scores rising to 91.66% on LEVIR-CD and 88.53% on AUUCDD. The incorporation of the attention mechanism (w/ Attention) achieves $F1$-scores of 92.12% and 89.08% on the LEVIR-CD and AUUCDD datasets, respectively. The full model achieves optimal performance, yielding $F1$-scores of 92.89% on LEVIR-CD and 90.17% on AUUCDD.

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TABLE IV

ABLATION STUDIES OF THE PROPOSED SGFM. FC: FEATURE CONCATENATION, SG: SEMANTIC GUIDANCE, EW: ENTROPY WEIGHT, AND MM: MODULATION MAP (%)

| Method | FC | SG | EW | MM | LEVIR-CD | | | AUUCDD | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | F1↑ | IoU↑ | Rec↑ | F1↑ | IoU↑ | Rec↑ |
| Baseline | | | | | 90.21 | 83.94 | 89.42 | 87.21 | 79.09 | 85.86 |
| w/ Feature Concat | ✓ | | | | 90.56 | 84.26 | 89.89 | 87.53 | 79.63 | 86.12 |
| w/ Semantic Guide | | ✓ | | | 91.32 | 84.98 | 90.21 | 88.19 | 80.18 | 86.84 |
| w/ Entropy Weight | | ✓ | ✓ | | 92.25 | 85.76 | 90.89 | 89.31 | 81.52 | 87.86 |
| Full Model | | ✓ | ✓ | ✓ | **92.89** | **86.17** | **91.88** | **90.17** | **82.34** | **89.13** |

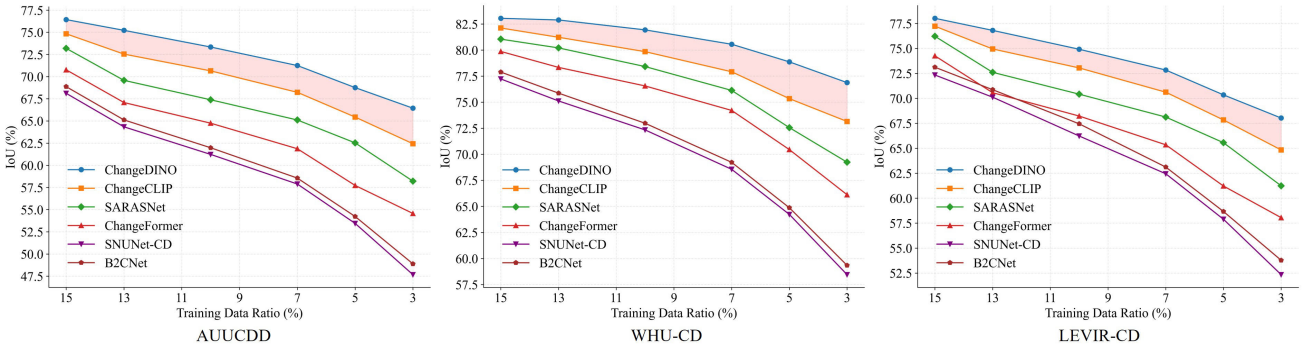



Fig. 11. Data-scarce performance comparison on three datasets.

TABLE V

ABLATION STUDIES OF PEM (%)

| Variants | LEVIR-CD | | | AUUCDD | | |
|---|---|---|---|---|---|---|
| | F1↑ | IoU↑ | Rec↑ | F1↑ | IoU↑ | Rec↑ |
| Baseline | 91.32 | 84.66 | 89.82 | 87.64 | 79.31 | 86.08 |
| w/ $L^t_{s,i}$ | 91.66 | 84.94 | 90.20 | 88.53 | 80.19 | 87.08 |
| w/ Attention | 92.12 | 85.66 | 90.87 | 89.08 | 81.08 | 87.84 |
| Full Model | **92.89** | **86.17** | **91.88** | **90.17** | **82.34** | **89.13** |

### *E. Data-Scarce Performance Analysis*

To comprehensively evaluate the robustness and practical applicability of ChangeDINO in real-world scenarios where labeled data is limited, we conduct extensive experiments under data-scarce conditions. We systematically vary the training data ratio from 15% to 3% across all three datasets, comparing ChangeDINO against strong baselines including ChangeCLIP, ChangeFormer, and SARASNet, all implemented with ResNet50 backbone for fair comparison.

As illustrated in Fig. 11, ChangeDINO demonstrates exceptional resilience to training data reduction across all datasets. While all methods exhibit performance degradation as training data decreases, ChangeDINO maintains significantly superior performance margins. On the challenging AUUCDD dataset, ChangeDINO preserves approximately 66% IoU performance even with only 3% training data, substantially outperforming ChangeCLIP (62%), SARASNet (58%), and ChangeFormer (55%). This represents a 4–12 percentage point advantage over competing methods under extreme data scarcity. Notably, efficiency-oriented methods such as SNUNet-CD and B2CNet exhibit more pronounced performance degradation under data-scarce conditions, further demonstrating the advantage of leveraging rich pretrained representations from Grounding DINO.

The degradation curves reveal distinctive behavioral patterns that highlight ChangeDINO's architectural advantages. Unlike conventional methods that exhibit steep performance drops, ChangeDINO demonstrates a more gradual and controlled decline. Specifically, when training data is reduced from 15% to 9%, ChangeDINO's average performance across all datasets decreases by merely 2%, while ChangeCLIP, ChangeFormer, and SARASNet experience more substantial drops of approximately 3%, 4%, and 3.5%, respectively. This graceful degradation pattern indicates superior generalization capabilities and reduced overfitting to limited training samples.

The superior performance under data scarcity can be attributed to the rich prior knowledge embedded in Grounding DINO's pretrained representations. Unlike ChangeCLIP, which relies on CLIP's global image–text contrastive learning features that lack specific object localization capabilities, ChangeDINO leverages Grounding DINO's sophisticated object detection and grounding capabilities. This enables the model to transfer fine-grained spatial understanding and semantic knowledge from large-scale pretraining, significantly reducing dependence on task-specific training data.

These results collectively demonstrate that the additional computational overhead introduced by Grounding DINO integration is well-justified from a practical deployment perspective. In real-world UAV monitoring scenarios where annotation resources are limited, the ability to maintain robust performance under data-scarce conditions represents a more

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TABLE VI
COMPUTATIONAL COMPLEXITY AND EFFICIENCY ANALYSIS ON LEVIR-CD DATASET

| Method | Backbone | Trainable Params (M) | GFLOPs | Training Time/Epoch | 256×256 | | 512×512 | | 1024×1024 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | FPS | Mem(GB) | FPS | Mem(GB) | FPS | Mem(GB) |
| SNUNet-CD | – | 18.8 | 18.3 | 7.7 min | 95 | 2.6 | 38 | 4.8 | 15 | 7.2 |
| B2CNet | ResNet18 | 16.1 | 22.4 | 8.6 min | 89 | 3.0 | 34 | 5.4 | 13 | 7.8 |
| ChangeFormer | ViT-B | 86.2 | 22.4 | 15.3 min | 42 | 4.8 | 17 | 7.2 | 5 | 10.6 |
| ChangeMamba | VMamba-S | 49.9 | 114.8 | 13.5 min | 58 | 4.2 | 23 | 6.8 | 7 | 9.8 |
| ChangeCLIP | ResNet50 | 35.9 | 15.4 | 8.7 min | 84 | 3.2 | 36 | 5.6 | 12 | 8.2 |
| ChangeCLIP | ViT-B | 87.4 | 28.6 | 16.8 min | 38 | 5.6 | 15 | 8.4 | 6 | 11.6 |
| ChangeDINO | ResNet50 | 38.2 | 28.9 | 11.9 min | 68 | 4.5 | 32 | 6.2 | 11 | 8.8 |
| ChangeDINO | ViT-B | 91.8 | 35.2 | 18.4 min | 32 | 6.8 | 15 | 8.8 | 6 | 11.9 |

TABLE VII
ABLATION STUDY ON TEXT PROMPT SELECTION AND PRETRAINING IMPACT

| Settings | LEVIR-CD | | | | WHU-CD | | | |
|---|---|---|---|---|---|---|---|---|
| | Recall↑ | F1↑ | IoU↑ | ΔF1 | Recall↑ | F1↑ | IoU↑ | ΔF1 |
| w/ Weights + "buildings" | 91.88 | 92.89 | 86.17 | - | 94.79 | 95.79 | 90.01 | - |
| w/ Weights + "construction" | 91.73 | 92.84 | 86.09 | -0.05 | 94.61 | 95.63 | 90.06 | -0.16 |
| w/ Weights + "structure" | 91.65 | 92.76 | 85.98 | -0.13 | 94.52 | 95.58 | 89.98 | -0.21 |
| w/ Weights + "object" | 90.89 | 91.12 | 83.87 | -1.77 | 93.12 | 94.23 | 89.12 | -1.56 |
| w/ Weights + "vegetation" | 88.56 | 89.67 | 81.34 | -3.22 | 90.89 | 92.34 | 85.89 | -3.45 |
| w/o Weights + "buildings" | 82.14 | 84.67 | 73.36 | -8.22 | 84.60 | 86.87 | 72.40 | -8.92 |

critical advantage than marginal gains on fully supervised benchmarks.

### F. Computational Complexity Analysis

We provide a comprehensive analysis of ChangeDINO's computational complexity on the LEVIR-CD dataset. All experiments are conducted on NVIDIA A100 GPU with consistent settings—batch size 1, input size 256 × 256—to ensure fair comparison as shown in Table VI.

ChangeDINO, built on ResNet50, requires only 38.2 M trainable parameters, just 6.4% more than ChangeCLIP (35.9 M), demonstrating efficient parameter utilization despite the integration of Grounding DINO features. While the GFLOPs increase from 15.4 to 28.9, the additional computation enhances representational capacity, yielding an $F1$-score of 92.89% compared to 92.01% on LEVIR-CD. Moreover, ChangeDINO achieves an inference speed of 68 FPS, maintaining competitive speed with other advanced approaches and remaining suitable for real-time applications.

With a GPU memory usage of 4.5 GB, ChangeDINO remains within practical limits for modern hardware, facilitating deployment on standard deep learning platforms. Training time per epoch of 11.9 min is reasonable and comparable to other vision–language methods, demonstrating the method's practical feasibility. Since the Grounding DINO backbone is frozen, the additional GFLOPs arise only from forward-pass feature extraction, not backpropagation, thereby keeping the increase in training time marginal (8.7–11.9 min per epoch), while lightweight fusion modules SGFM + PEM add minimal computational overhead. Compared with efficiency-oriented methods such as SNUNet-CD and B2CNet, ChangeDINO achieves consistently superior performance across all datasets despite moderate additional overhead, demonstrating a favorable efficiency–accuracy tradeoff. Furthermore, under higher input resolutions representative of real-world UAV deployment scenarios, ChangeDINO maintains an acceptable memory footprint and inference speed across all tested resolutions, demonstrating stable scalability for practical applications.

### G. Text Prompt Selection Strategy and Pretraining Analysis

To address concerns regarding the text prompt selection strategy and the importance of Grounding DINO's pretraining, we conduct comprehensive ablation studies on both LEVIR-CD and WHU-CD datasets. The results are presented in Table VII.

*1) Text Prompt Selection:* We evaluate five different prompts for change detection: "buildings," "construction," "structure," "object," and "vegetation." The experimental results demonstrate that building-related prompts ("buildings," "construction," "structure") yield similar performance when using pretrained Grounding DINO weights. Specifically, on the LEVIR-CD dataset, the $F1$-score variations are minimal—92.89% versus 92.84% versus 92.76%—while on the WHU-CD dataset, the differences are equally negligible—95.79% versus 95.63% versus 95.58%. This robustness indicates that Grounding DINO's rich semantic representations can effectively capture building-related concepts regardless of the specific terminology used, reducing the sensitivity to prompt engineering.

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TABLE VIII
PERFORMANCE COMPARISON ON DIFFERENT LAND COVER TYPES FROM SECOND DATASET (%)

| Method | Backbone | Low Vegetation | Bare Soil |
|---|---|---|---|
| | | F1 / IoU / Rec | F1 / IoU / Rec |
| CGNet | VGG16 | 74.23 / 59.08 / 72.45 | 77.34 / 62.89 / 75.12 |
| ChangeCLIP | ResNet50 | 75.67 / 60.82 / 73.89 | 78.91 / 65.08 / 76.67 |
| ChangeDINO | ResNet50 | **77.34 / 63.02 / 75.56** | **80.67 / 67.63 / 78.45** |

However, when using semantically mismatched prompts, performance degrades significantly. Using the generic prompt "object" results in $F1$-scores of 91.12% on LEVIR-CD and 94.23% on WHU-CD, representing drops of 1.77% and 1.56%, respectively. More notably, using the semantically irrelevant prompt "vegetation" for building change detection leads to substantial performance degradation, with $F1$-scores dropping to 89.67% on LEVIR-CD and 92.34% on WHU-CD (decreases of 3.22% and 3.45%). These results demonstrate that while Grounding DINO exhibits robustness within semantically related prompts, task-relevant prompt selection remains important for optimal performance.

*2) Pretraining Importance:* The critical role of Grounding DINO's pretraining becomes evident when comparing results with and without pretrained weights. Without pretraining, the performance drops dramatically across both datasets: $F1$-scores decrease by 8.22% on LEVIR-CD—from 92.89% to 84.67%—and 8.92% on WHU-CD—from 95.79% to 86.87%. These substantial performance degradations highlight that the pretrained vision–language knowledge in Grounding DINO is essential for effective semantic guidance in change detection tasks.

### H. Generalization Analysis Across Different Change Types

To validate ChangeDINO's generalization capability on nonbuilding categories, we conduct experiments on the SECOND dataset. Specifically, we extract two semantic categories significantly different from buildings—low vegetation and bare soil—from SECOND's multiclass annotations and convert them into binary change detection tasks. For each task, we retain all image pairs from the dataset but only mark pixels of the target category as changed (1), with all other pixels marked as unchanged (0). This setting simulates practical monitoring scenarios for specific land cover types, where not all images contain changes of the target category. The dataset split follows SECOND's official configuration. In our experiments, we use text prompts "low vegetation" and "bare soil" to guide Grounding DINO in extracting corresponding semantic priors.

Table VIII presents the performance comparison between ChangeDINO and baseline methods on these two tasks. The experimental results demonstrate that ChangeDINO achieves optimal performance on both low vegetation and bare soil change detection. For the low vegetation task, ChangeDINO improves the $F1$-score by 1.67% compared to ChangeCLIP (77.34% versus 75.67%); for the bare soil task, the improvement is 1.76% (80.67% versus 78.91%). It is noteworthy that the overall performance on the bare soil task is slightly higher than that on the low vegetation task, which can be attributed to the fact that bare soil regions exhibit more distinct spectral differences in bi-temporal images, while vegetation changes are more affected by seasonal factors.

These experiments fully demonstrate that ChangeDINO not only excels in building change detection but also exhibits excellent generalization capability on other land cover types, showcasing the broad applicability of vision–language priors in RSCD.

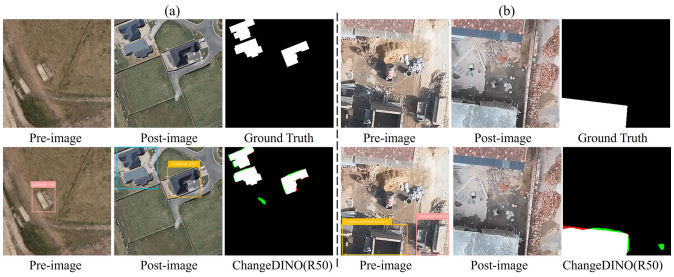



Fig. 12. Failure case of ChangeDINO: Grounding DINO generates a semantically incorrect detection ("buildings," confidence: 0.36) in the preimage. This confident but erroneous response is treated as an informative signal and amplified, ultimately leading to a false positive. (a) Visual morphology ambiguity; (b) Domain-specific prompt over-sensitivity.

### I. Failure Case Analysis

While SGFM's activation-based semantic guidance is effective in the vast majority of scenarios, it measures activation concentration as a proxy for semantic saliency rather than semantic correctness, meaning that erroneous but high-confidence VLM predictions may still be amplified into false positives. We systematically reviewed the test sets of all four datasets and recorded the observed failure cases: 1 of 2048 on LEVIR-CD, 2 of 744 on WHU-CD, 0 of 4000 on SYSU-CD, and 4 of 3000 on AUUCDD.

These failures fall into two categories. The first involves **visual morphology ambiguity**, where nontarget objects with geometric or structural similarity to the prompted category are incorrectly activated, producing isolated false positives in semantically stable regions. The second involves **domain-specific prompt over-sensitivity**, where fine-grained prompts trigger spurious responses on transient scene elements that superficially resemble the target category. Fig. 12 illustrates representative cases of both categories, where (a) corresponds to visual morphology ambiguity and (b) corresponds to domain-specific prompt over-sensitivity. Both failure types share the same root cause: SGFM relies solely on activation concentration as a reliability proxy and cannot verify semantic correctness, indicating that these represent inherent limitations of adapting open-vocabulary detectors to change detection. Potential strategies that could be explored in future work include adaptive confidence-aware gating to address visual morphology ambiguity, and causal-intervention verification to suppress spurious activations in construction-specific scenes.

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## V. Conclusion

In this article, we propose ChangeDINO, a RSCD framework that effectively integrates vision–language prior knowledge. By systematically addressing the structural challenges of adapting open-vocabulary detector priors to pixel-level prediction through a bi-level feature entanglement strategy—realized through our proposed SGFM and PEM, ChangeDINO achieves SOTA performance on standard RSCD benchmarks, LEVIR-CD and WHU-CD. To further evaluate its generalizability, we construct a novel drone-based dataset, AUUCDD—the first of its kind specifically designed for detecting unauthorized urban construction. ChangeDINO also achieves leading performance on this challenging dataset, demonstrating its practical value in low-altitude urban applications. Finally, we evaluate our method in data-scarce scenarios, further confirming the effectiveness of leveraging prior knowledge from a vision–language-based object detection model and its potential in situations with limited annotation resources. We hope this work provides a principled reference for adapting an open-vocabulary detector to dense prediction tasks. Furthermore, as AUUCDD is derived from an operational urban monitoring platform with continuous data collection, we plan to progressively expand this dataset in future work to enhance its scale and scene diversity, thereby strengthening its value as a benchmark for unauthorized construction detection.

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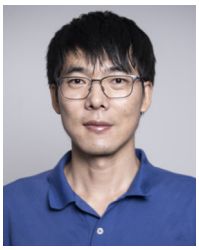
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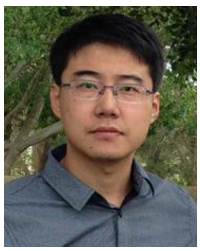
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